

BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

1. RINGKASAN GUNCANGAN TERKUAT

Stasiun GKJM _ 2026-01-27 06_14_05 UTC mencatat nilai percepatan tanah maksimum (PGA) sebesar 1.413859 m/s² atau 141.386 Gal pada komponen HNE. Berdasarkan klasifikasi SIG-BMKG yang digunakan oleh BSMA, hasil tersebut termasuk Skala III (KERUSAKAN RINGAN) dengan rentang intensitas VI.

Network	IA
Station	GKJM
Location	00
Sampling Rate	100.000 Hz
Start Time	2026-01-27T06:14:05.650000Z
End Time	2026-01-27T06:21:05.640000Z
Channels	HNE, HNN, HNZ

SKALA III SIG-BMKG | KERUSAKAN RINGAN

Warna: Kuning | Rentang intensitas: VI

Dampak: Bagian nonstruktur bangunan mengalami kerusakan ringan, seperti retak rambut pada dinding, atap bergeser ke bawah dan sebagian berjatuhan.

2. PARAMETER STRONG MOTION

Channel	PGA (Gal)	PGV (cm/s)	PGD (cm)	Arias (m/s)	D5-95 (s)	SIG
HNE	141.386	3.2315	0.1635	0.037806	3.25	III (VI)
HNN	66.310	1.4704	0.0619	0.029168	3.42	II (III-V)
HNZ	43.360	1.0179	0.0535	0.010252	4.79	II (III-V)

Parameter komponen terkuat:

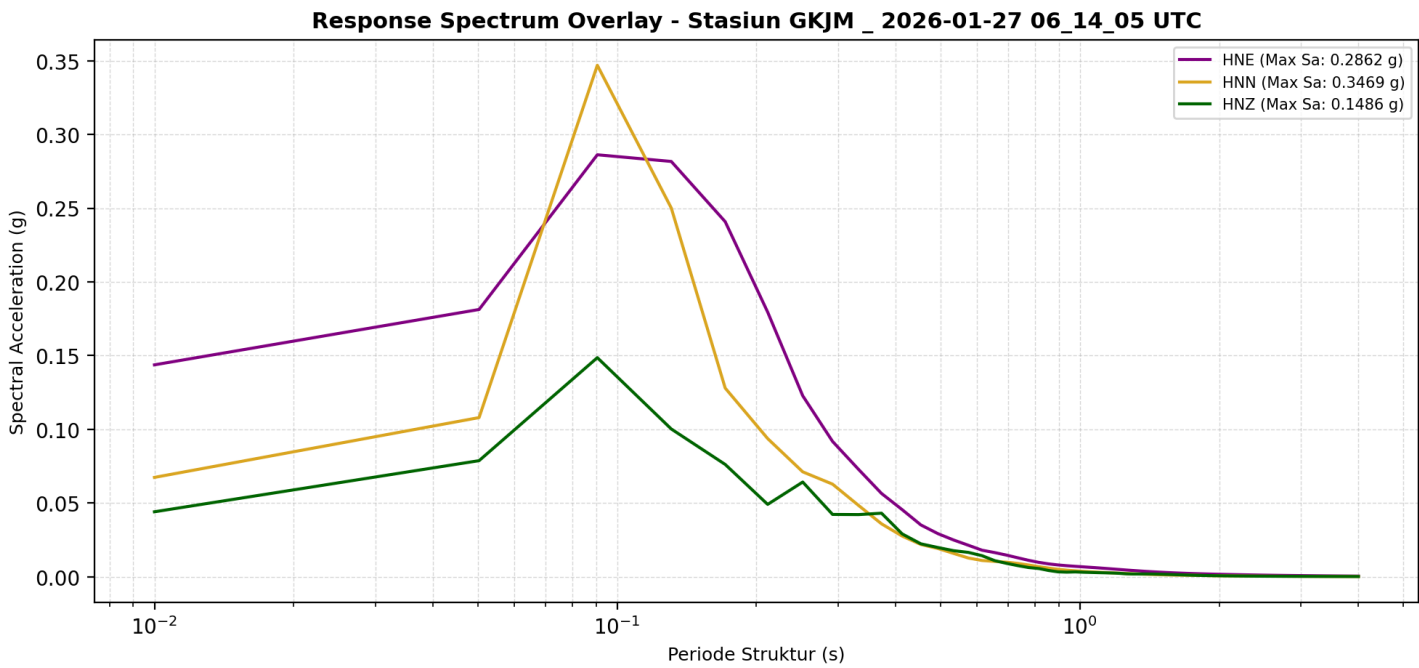
PGA = 1.413859 m/s² (141.386 Gal); PGV = 3.2315 cm/s; PGD = 0.1635 cm; Arias Intensity = 0.037806 m/s; D5-95 = 3.25 s.

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3. RESPONSE SPECTRUM

Kurva response spectrum menunjukkan percepatan spektral sebagai fungsi periode struktur berdasarkan hasil pemrosesan response spectrum pada pipeline BSMA.

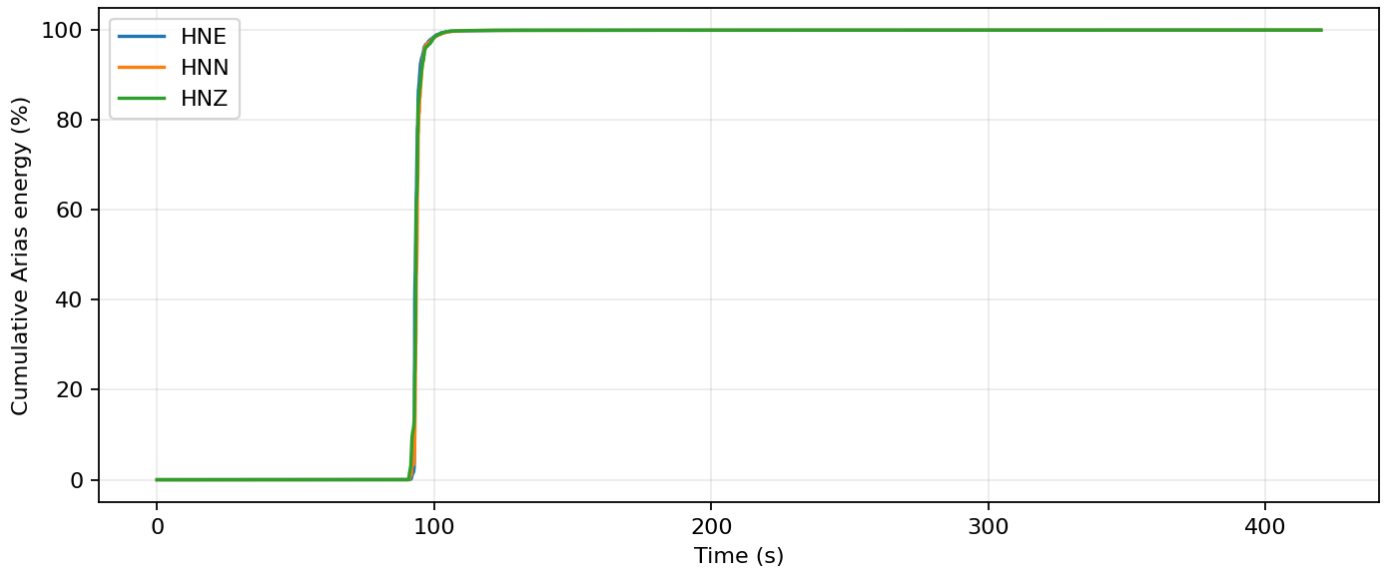


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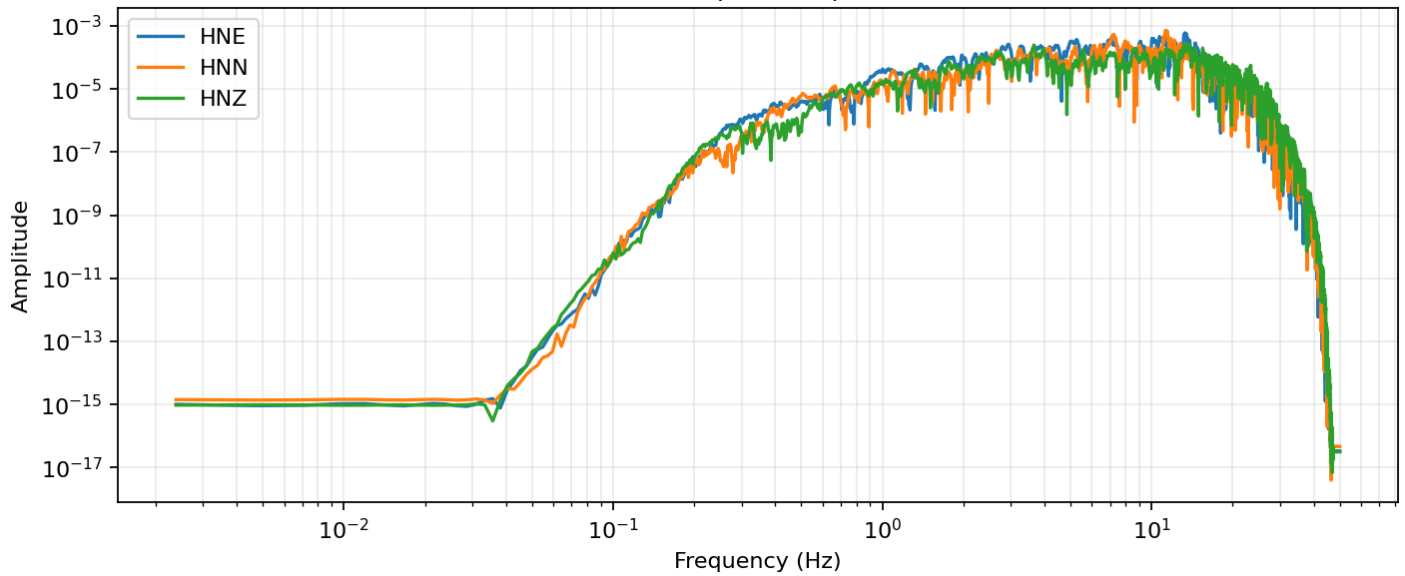
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4. HUSID PLOT DAN FOURIER AMPLITUDE SPECTRUM

Husid energy curve



Fourier amplitude spectrum



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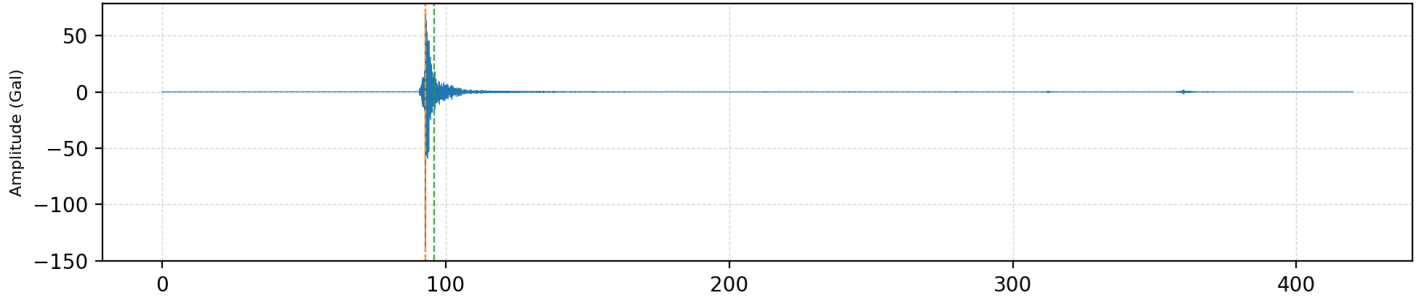
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5. WAVEFORM ANALYSIS - HNE

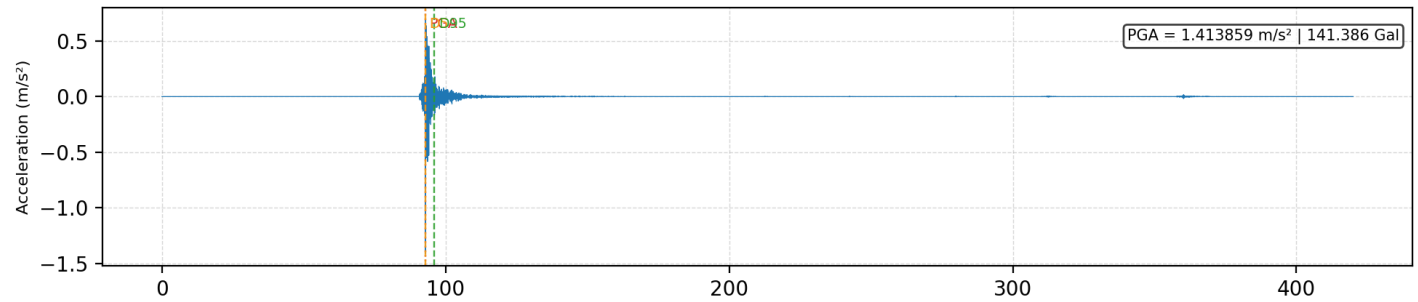
PGA: 141.386 Gal | PGV: 3.2315 cm/s | PGD: 0.1635 cm | D5-95: 3.25 s

Analisis Kinematik Ground Motion
Stasiun: GKJM _ 2026-01-27 06_14_05 UTC | Channel: HNE

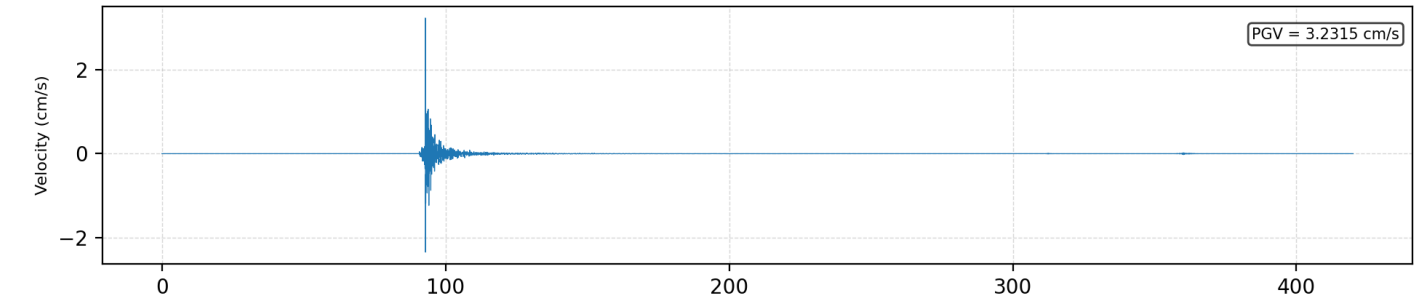
1. Raw Waveform - HNE



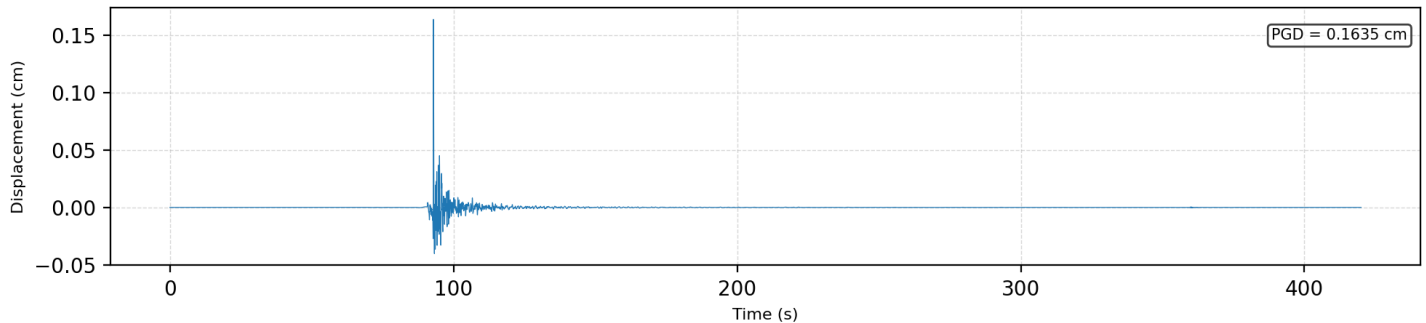
2. Filtered Ground Acceleration



3. Ground Velocity



4. Ground Displacement



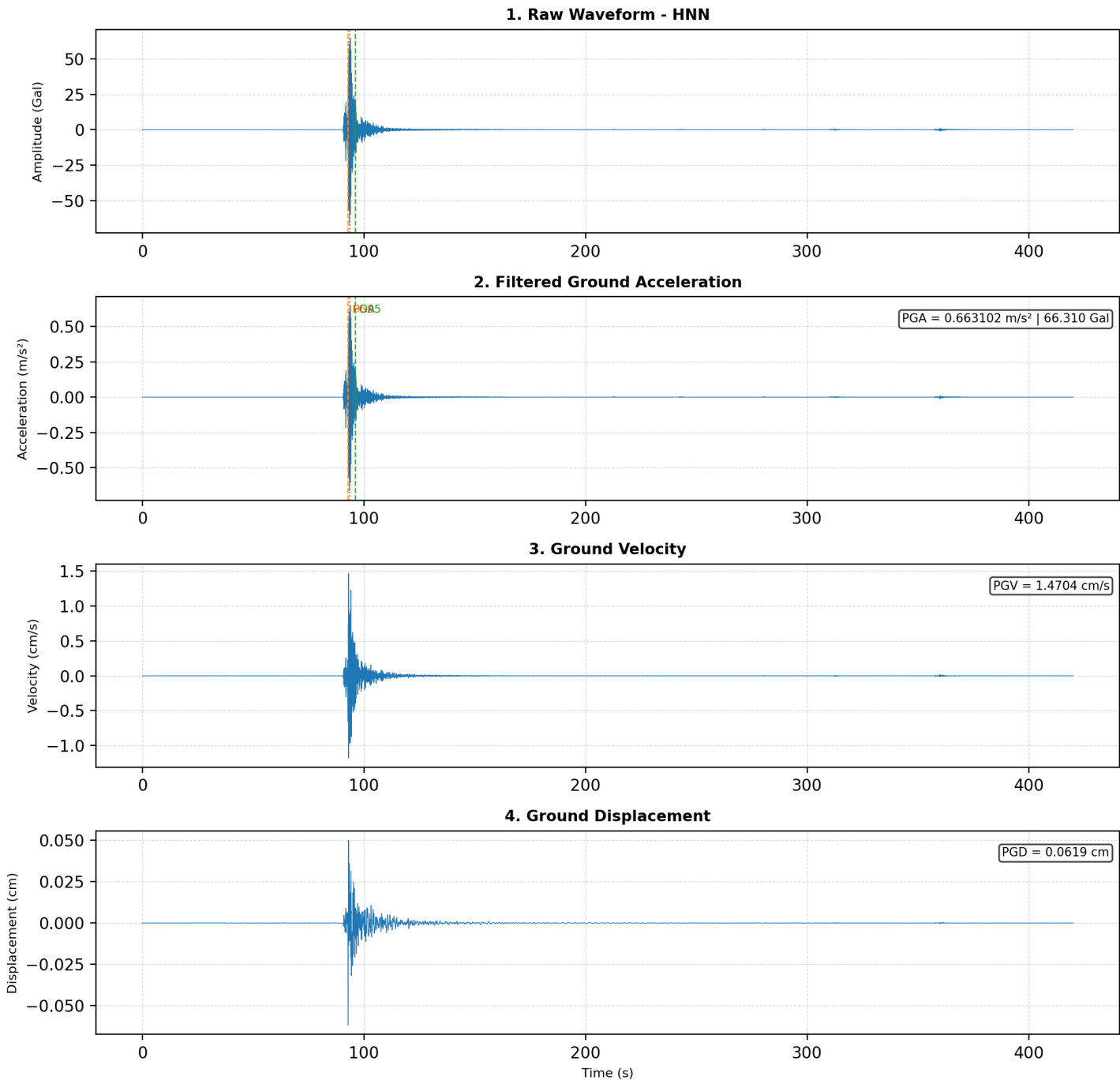
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5. WAVEFORM ANALYSIS - HNN

PGA: 66.310 Gal | PGV: 1.4704 cm/s | PGD: 0.0619 cm | D5-95: 3.42 s

Analisis Kinematik Ground Motion
Stasiun: GKJM _ 2026-01-27 06_14_05 UTC | Channel: HNN



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5. WAVEFORM ANALYSIS - HNZ

PGA: 43.360 Gal | PGV: 1.0179 cm/s | PGD: 0.0535 cm | D5-95: 4.79 s

Analisis Kinematik Ground Motion
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